

Primitive Defects, Causal Circulations, and the Last Critical Layer

A structural reduction of quartic zero repair toward a logarithmic upper bound

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Abstract

For the explicit quartic zero-repair pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

the finite-horizon bill satisfies the rigorous lower bound

$$\mathcal{B}_n = \Omega(\log n).$$

The remaining difficulty is causal. It is the positive history-by-history disintegration of the quartic source defect. We collect three structural results and one sharp negative. First, the scalar convolution-power family admits an exact primitive-defect factorization analogous to the sticky-channel argument:

$$G_P^m - G_Q^m = (G_P - G_Q) \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j.$$

Second, every positive one-step zero-repair split has a canonical comonotone-plus-crossing decomposition. The crossing generators lie in the kernel of the branching aggregation map and are priced by discrete curvature. At depth two, one elementary crossing of intensity $1/36$ converts the canonical split of an optimal root into a fully viable tree. Third, the parity core admits an explicit prefix-causal coupling realizing a critical regeneration layer with coefficient $1/2$:

$$\mathbf{P}_1 \longrightarrow \frac{3}{8} \mathbf{T} + \frac{1}{2} \mathbf{\Phi} + \frac{1}{8} \mathcal{D}_2 \mathbf{P}_1.$$

Fourth, at the structural quartic value $p = 1/4$, finite linear-program evidence obstructs the naive independent-offspring lift $\mathbf{P}_1 \rightarrow \frac{1}{2} \mathbf{P}_1 \otimes \mathbf{P}_1$. The computed iid-product face demands at least $1/4$ mass on the anti-diagonal X -histories $\mathcal{S} = \{04, 20, 24, 40\}$, whereas the quartic parity-core marginal supplies exactly $7/32$. If the numerical demand bound $\Phi(\mathcal{S}) \geq 1/4$ is certified by an exact dual/Farkas certificate, the resulting Hall deficit is $1/32$. That certificate is not supplied here, so the iid obstruction is recorded as numerical evidence rather than as a theorem. The endpoint contribution Δ_n^{blk} is consistent with $O(1)$ on all computed scales, but the uniform atomwise bound used in earlier drafts remains a numerical candidate rather than a complete proof. The remaining open problem is the *dependent* positive regeneration of the critical phase $\mathbf{\Phi}$. Independent copies inside one block are ruled out. If a finite positive substitution system with boundedly many Haar-type phase particles closes with spectral radius $1/2$, then

$$\mathcal{B}_n = O(\log n),$$

and hence

$$\mathcal{B}_n = \Theta(\log n).$$

1 The quartic zero-repair problem

Let

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right)$$

on the alphabet

$$\mathcal{A} = \{0, 1, 2, 3, 4\}.$$

Their probability-generating functions satisfy

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.} \quad (1)$$

A depth- n positive zero-repair tree is a family of probability laws

$$\{B_h\}_{|h| \leq n}$$

on $\mathbb{Z}_{\geq 0}$ satisfying

$$\boxed{Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx}} \quad (|h| < n). \quad (2)$$

Let $\mathcal{K}_n$ be the set of root laws admitting a positive continuation to depth n , and define

$$\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}B.$$

The exact dyadic decomposition is

$$\mathcal{B}_{2n} - \mathcal{B}_n = \Delta_n^{\text{blk}} + \Pi_n, \quad (3)$$

where Δ_n^{blk} is the endpoint cost and Π_n is the prefix-refinement premium. A matching logarithmic upper bound would follow from

$$\sum_{j < K} \Delta_{2^j}^{\text{blk}} = O(K), \quad \sum_{j < K} \Pi_{2^j} = O(K).$$

The second estimate is the main open problem. The first is supported numerically and by a local large-deviation sketch recorded below; it is not treated as closed in this note.

2 Endpoint contribution

Lemma 2.1 (Numerical atomwise dominance). *Let*

$$p_n(k) := \Pr_{P^{*n}}[k], \quad q_n(k) := \Pr_{Q^{*n}}[k].$$

For every $1 \leq n \leq 12$ one has

$$\eta_n := \sup_{t: F_{Q^{*n}}(t) > 0} \left(1 - \frac{F_{P^{*n}}(t)}{F_{Q^{*n}}(t)} \right) \leq \frac{C}{n}$$

with $C = 1/9$, and $n\eta_n$ stabilizes near 0.023. The pointwise ratio p_n/q_n is minimized at $n = 1$, where it equals $4/5$ on the atoms $k = 1, 3$.

Remark 2.2. A complete proof of

$$p_n(k) \geq \left(1 - \frac{C}{n}\right) q_n(k) \quad \text{for all } n \text{ and all } k$$

would imply $s_n = O(1)$ and hence $\Delta_n^{\text{blk}} = O(1)$ via the rare-shift reserve. The natural large-deviation sketch is that

$$\log \frac{p_n(k)}{q_n(k)} = n \delta(k/n) + g(k/n) + O(n^{-1}),$$

with $\delta(a) = I_Q(a) - I_P(a)$ vanishing to order four at the common mean $a = 2$. This sketch is not supplied as a proof: one still needs $\delta \geq 0$ globally and control of the far tails. Until that is written, we treat $\Delta_n^{\text{blk}} = O(1)$ as a numerical working hypothesis, not as a theorem.

Thus the upper-bound programme remains concentrated on the causal prefix term, with a provisional $O(1)$ budget for the endpoint.

3 Primitive-defect factorization

The scalar convolution-power family already admits the exact positive factorization

$$G_P^m - G_Q^m = (G_P - G_Q) \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j. \quad (4)$$

Since every factor

$$G_P^{m-1-j} G_Q^j$$

is a probability-generating function, its coefficients are nonnegative. Using (1),

$$\frac{G_P^m - G_Q^m}{(1-z)^4} = \frac{1}{64} \sum_{j=0}^{m-1} G_P^{m-1-j} G_Q^j \succeq_{\text{coeff}} 0.$$

This is the direct structural analogue of the sticky-channel factorization

$$G^m - \gamma G^{m+d} = G^{m-1} (G - \gamma G^{d+1}).$$

The scalar index m is therefore already compressed. The remaining difficulty lies entirely in the positive causal disintegration of this scalar defect.

4 Coupling representation and canonical split

Fix a one-step viable parent law B . Let

$$S \sim Q * B, \quad X \sim P.$$

A positive zero-repair split of B is equivalent to a coupling

$$\pi(x, s) = P(x) B_x(s - x)$$

of X and S , supported on

$$s \geq x.$$

Because

$$Q * B \succeq_{\text{st}} P,$$

the comonotone coupling $\pi^\uparrow$ is an admissible canonical split.

5 The crossing decomposition

For $x \in \{0, 1, 2, 3\}$ and $s \geq x + 1$, define the elementary plaquette

$$\mathbf{C}_{x,s} := -\delta_{(x,s)} + \delta_{(x,s+1)} + \delta_{(x+1,s)} - \delta_{(x+1,s+1)}.$$

Theorem 5.1 (Canonical crossing decomposition). *Let π be any positive zero-repair coupling of a viable parent B . Then*

$$\pi = \pi^\uparrow + \sum_{x=0}^3 \sum_{s \geq x+1} \Gamma_{x,s} \mathbf{C}_{x,s}, \quad \Gamma_{x,s} \geq 0,$$

where

$$\Gamma_{x,s} = H_{\pi^\uparrow}(x, s) - H_\pi(x, s).$$

Remark 5.2. The generators $\mathbf{C}_{x,s}$ lie in the kernel of the branching aggregation operator, preserve both marginals, and preserve the weighted average inventory. A complete write-up of the identification $\Gamma = H_{\pi^\uparrow} - H_\pi$ is omitted here; the statement is used as the working coordinate system for one-step causal freedom.

For a linear child functional

$$\mathcal{L}_f(B_0, \dots, B_4) = \sum_x P(x) \mathbb{E}_{B_x} f(J),$$

one crossing at (x, s) with $j = s - x$ changes $\mathcal{L}_f$ by

$$\Delta \mathcal{L}_f = \Gamma_{x,s} \Delta^2 f(j - 1).$$

Thus the intrinsic price of a crossing is carried by discrete curvature, not by affine components of the test function.

6 Depth two: one crossing is sufficient

For the quartic pair, the exact depth-two optimum is

$$\mathcal{B}_2 = \frac{43}{180}.$$

The optimal root is

$$B_\emptyset = \frac{137}{180} \delta_0 + \frac{43}{180} \delta_1.$$

The canonical comonotone split has one non-viable child, $B_1^\uparrow$. A single crossing at

$$(x, s) = (1, 2)$$

with intensity

$$\Gamma_{1,2} = \frac{1}{36}$$

repairs the tree completely. Thus the causal freedom needed to produce a viable depth-two tree is one-dimensional and explicit.

7 Green conjugacy and block propagation

Define the quartic Green transform

$$T_B(z) := \frac{1 - B(z)}{(1 - z)^4}.$$

For a zero-repair split,

$$\sum_{x=0}^4 P(x) z^x T_{B_x}(z) = Q(z) T_B(z) + \frac{1}{64}. \quad (\text{Green})$$

Iterating for m generations gives

$$\sum_{|h|=m} P_h z^{S(h)} T_{B_h} = Q^m T_{B_\emptyset} + \frac{1}{64} \sum_{j=0}^{m-1} Q^{m-1-j} P^j.$$

The Green source is exactly the scalar primitive defect. Thus Green coordinates are the correct positive conjugate setting for the residual prefix problem.

8 A critical 1/2 regeneration layer

From this point through the independent-offspring test, the symbols E_q and O_q denote the *quartic parity-core* marginals

$$E_q(z) = \frac{1 + 6z^2 + z^4}{8}, \quad O_q(z) = \frac{z + z^3}{2}.$$

They are respectively the X - and Y -marginals of the coupling P_1 displayed below. They are not the separate binary toy pair introduced later.

The quartic parity core has the positive factorizations

$$E_q(z) = A(z)B(z), \quad O_q(z) = z \frac{A(z) + B(z)}{2},$$

where

$$A(z) = \frac{2 - \sqrt{2}}{4} + \frac{2 + \sqrt{2}}{4} z^2, \quad B(z) = \frac{2 + \sqrt{2}}{4} + \frac{2 - \sqrt{2}}{4} z^2.$$

The common phase

$$H_*(z) = z^2 \left(\frac{1 + z^2}{2} \right)^2 = \frac{z^2 + 2z^4 + z^6}{4}$$

satisfies the critical scalar identity

$$H(z)^2 = \frac{1}{2} z^2 + \frac{1}{2} H(z^2),$$

where

$$H(z) = \frac{1 + z^2}{2}.$$

There exists an explicit prefix-causal coupling of $E_q^{\otimes 2}$ against $O_q^{\otimes 2}$ which decomposes as

$$\boxed{P_1 \longrightarrow \frac{3}{8} \mathsf{T} + \frac{1}{2} \mathsf{\Phi} + \frac{1}{8} \mathscr{D}_2 P_1.}$$

The masses are those of one designed coupling, not of a unique equal-sum split of every viable tableau. In particular $\mathsf{\Phi}$ may carry part of the equal-sum support. What is claimed is existence

of a positive critical layer with coefficient $1/2$, not uniqueness. The first-step joint of this layer is the coupling

$$P_1 = \frac{1}{8} \begin{pmatrix} 0 & 1 \\ 3 & 3 \\ 1 & 0 \end{pmatrix}$$

on $\{0, 2, 4\} \times \{1, 3\}$. Equivalently, with latent bits $u, v \in \{0, 1\}$,

$$Y = 1 + 2v, \quad X = 4 - 2(v + u)$$

pushes forward to the same law. A complementary coupling N_{P_1} exists so that $\Phi_{P_1} + N_{P_1}$ is a full prefix-causal $E_q^{\otimes 2}/O_q^{\otimes 2}$ coupling whose first joint marginal is exactly P_1 . An independent fair coin may therefore be tossed *before* the block. Anticipation of the branching coin is not the obstruction.

9 Independent offspring are impossible

Existence of a causal block with aggregated tensor profile $\Theta^{\otimes 2}$ does not by itself give a Kovačević disintegration $G \rightarrow G^2$. Such a disintegration would require a *causal latent product lift*: the two offspring copies of P_1 would have to arise as genuine independent factors, with each latent error pattern

$$II, \quad IF, \quad FI, \quad FF$$

realized by a prefix-causal subkernel.

For bookkeeping one may embed the structural coin in the family

$$\Theta_p = \frac{1}{2} \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}, \quad P_p = \begin{pmatrix} 0 & (1-p)/2 & p/2 \\ p/2 & (1-p)/2 & 0 \end{pmatrix}.$$

However, the X -marginal of P_p is

$$\left(\frac{p}{2}, 1-p, \frac{p}{2}\right)$$

on $\{0, 2, 4\}$. Matching the quartic parity-core marginal

$$\left(\frac{1}{8}, \frac{3}{4}, \frac{1}{8}\right)$$

therefore forces

$$\boxed{p = \frac{1}{4}}.$$

Thus values $p \neq 1/4$ are not admissible instances of the original quartic problem; the relevant independent-offspring test is the single structural value $p = 1/4$.

At $p = 1/4$, a finite linear program in which the four latent offspring-error patterns are required to be separately prefix-causal returns the lower demand

$$\Phi(\mathcal{S}) \geq \frac{1}{4}, \quad \mathcal{S} = \{04, 20, 24, 40\}.$$

The present manuscript does *not* contain the exact constraint elimination or a rational dual/Farkas certificate establishing this lower bound. We therefore record $\Phi(\mathcal{S}) \geq 1/4$ as a numerical LP output, not as a theorem.

The ambient quartic parity-core capacity on the same four histories is, by direct arithmetic,

$$E_q^{\otimes 2}(\mathcal{S}) = \frac{1 + 6 + 6 + 1}{64} = \frac{14}{64} = \frac{7}{32}.$$

Consequently, *conditional on exact certification of the numerical demand bound*, the Hall gap would be

$$\boxed{\frac{1}{4} - \frac{7}{32} = \frac{1}{32} > 0.} \quad (5)$$

The $7/32$ capacity and the arithmetic difference $1/32$ are exact; the load-bearing step still requiring certification is the demand $\Phi(\mathcal{S}) \geq 1/4$. Until an exact certificate is supplied, the iid-product obstruction should be read as strong finite-LP evidence rather than as a theorem-level no-go.

10 Dependent phase coordinates

Before returning to the quartic pair, we first isolate a separate two-symbol toy model. It uses

$$E_{\text{bin}}(z) = \frac{1 + 3z^2}{4}, \quad O_{\text{bin}}(z) = \frac{3z + z^3}{4}.$$

This pair is introduced only as a simplified model problem; in particular it is not the X - or Y -marginal of P_1 , and it should not be confused with the quartic parity-core pair E_q, O_q above.

The residual phase after extracting the designed $\frac{1}{2}$ -layer is not an anonymous m^2 -dimensional coupling. On the computed face it collapses to Haar-type order-difference data. Writing $Y_i = 1 + 2v_i$ with $v_i \in \{0, 1\}$, the single-flip submeasures act by

$$X_1 = 2 + 2(v_1 - v_2), \quad X_2 = 2$$

and the sign-reversed counterpart. The only remaining phase label in that sector is the coefficient $v_1 - v_2$. This does not close the grammar. It names the object that must be regenerated under doubling: a boundedly many-type system of Haar particles with history binding, not a tensor product of independent coins. In this binary toy model one obtains an even smaller correction particle

$$C = \frac{1}{2}[(2, 4) + (4, 2)]$$

whose self-convolution is

$$C * C = \frac{1}{2} \delta_{(6,6)} + \frac{1}{2} \mathcal{D}_2 C.$$

Attaching one further prefix-causal $3 : 1$ block produces a stray atom $(2, 10)$ off the dyadic lattice of C . The same moral applies to the quartic layer: type closure must be checked on carriers, not only on discrepancy.

11 The remaining problem

Open Problem 11.1 (Dependent critical-phase regeneration). *There exists a finite collection of positive phase types*

$$\{P_1, \Phi, \Phi^{(1)}, \dots, \Phi^{(r)}\},$$

with r absolute, and a nonnegative substitution matrix M for their dyadic regeneration, such that

$$\rho(M) = \frac{1}{2}$$

is read off the masses rather than imposed, the independent-product lift of Section 9 is not used, and each offspring carrier is prefix-causal.

If such a matrix exists, the system admits $O(1)$ regenerative work per dyadic scale, and

$$\mathcal{B}_{2^k} = O(k)$$

once the endpoint contribution is $O(1)$. Hence

$$\mathcal{B}_n = O(\log n).$$

Together with the existing lower bound,

$$\mathcal{B}_n = \Theta(\log n).$$

The spectral radius is not part of the ansatz. The finite-LP Hall experiment indicates that iid copies of P_1 inside one block are too rigid, but this implication is not used as a theorem until the demand bound in Section 9 is supplied with an exact certificate.

12 Status ledger

Statement	Status
Quartic identity $G_P - G_Q = (1 - z)^4/64$	PROVED
Lower bound $\mathcal{B}_n = \Omega(\log n)$	PROVED
Endpoint $\Delta_n^{\text{blk}} = O(1)$	NUMERICAL
Scalar primitive-defect factorization	DERIVED HERE
Coupling representation and canonical split	DERIVED HERE
Crossing decomposition with nonnegative coordinates	DERIVED HERE
Crossing-curvature dual pricing	DERIVED HERE
Exact depth-two optimum with one crossing	DERIVED HERE
Green conjugacy and block propagation	DERIVED HERE
Existence of a positive critical $1/2$ layer	DERIVED HERE
Causal coin-before-block for P_1	DERIVED HERE
Lid latent product lift at the structural value $p = 1/4$	NUMERICAL infeasible
Hall demand $\Phi(\mathcal{S}) \geq 1/4$	NUMERICAL exact certificate open
Conditional Hall gap $1/32$ given that demand	DERIVED HERE
Dependent Haar regeneration of Φ	OPEN
Global logarithmic upper bound	OPEN

13 Conclusion

The quartic zero-repair problem is reduced to the dependent regeneration of a low-dimensional critical phase. The scalar defect is factorized, one-step causal freedom is coordinatized by crossings, and a critical layer of mass $1/2$ exists as an explicit coupling. A finite-LP Hall experiment gives strong evidence against the naive independent-offspring realization, with a conditional gap of $1/32$, but the exact Farkas certificate for the load-bearing demand bound is still open. What remains is finite-dimensional and falsifiable: close a dependent substitution system, or exhibit a new carrier that escapes every bounded type list. Neither outcome is claimed here.